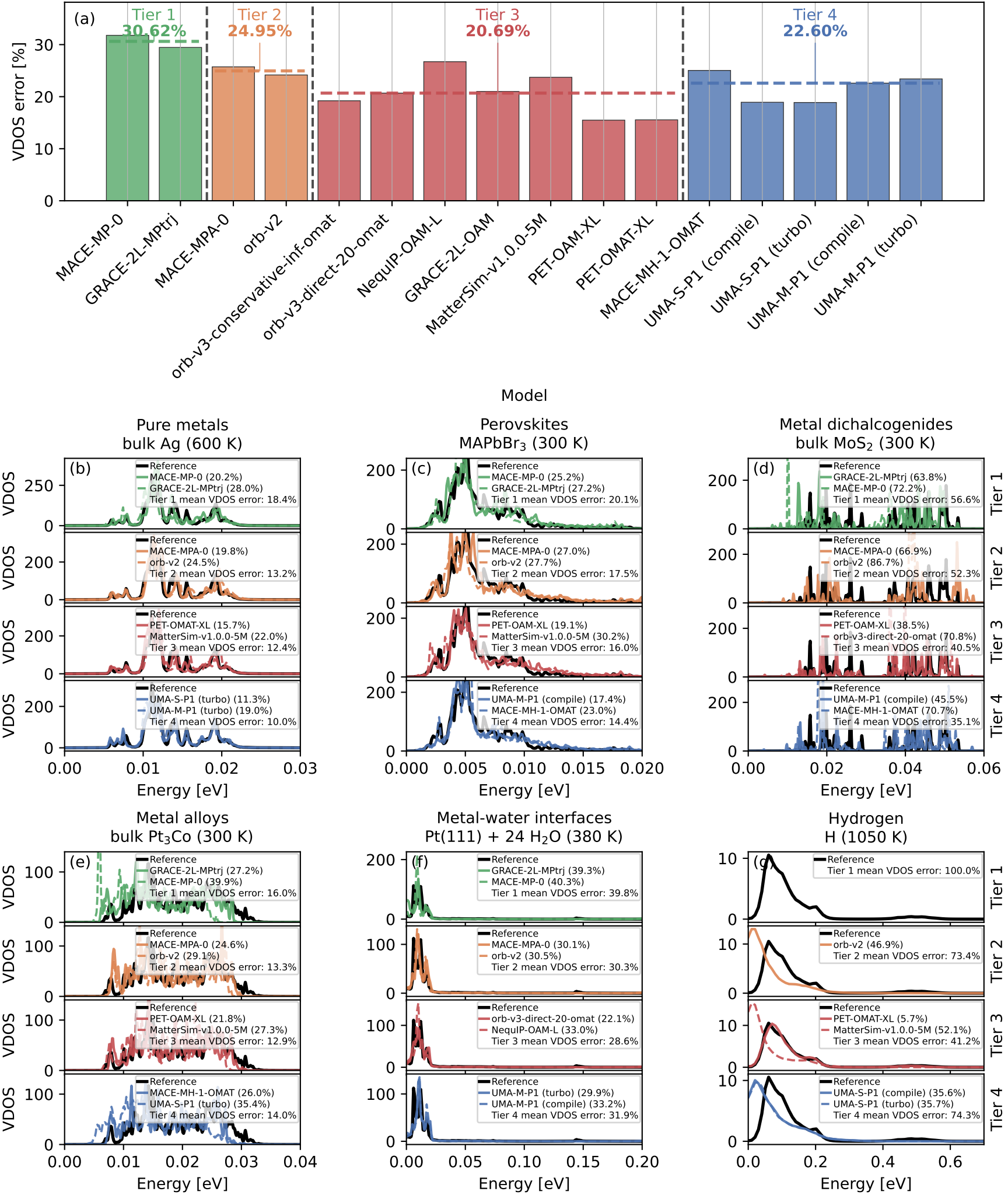

(a)
Tier 1
30.62%
Tier 2
24.95%
Tier 3
20.69%
Tier 4
22.60%
VDOS error [%]
MACE-MP-0
GRACE-2L-MPtrj
MACE-MPA-0
orb-v2
orb-v3-conservative-inf-omat
orb-v3-direct-20-omat
NequIP-OAM-L
GRACE-2L-OAM
MatterSim-v1.0.0-5M
PET-OAM-XL
PET-OMAT-XL
MACE-MH-1-OMAT
UMA-S-P1 (compile)
UMA-S-P1 (turbo)
UMA-M-P1 (compile)
UMA-M-P1 (turbo)
Model
Pure metals
bulk Ag (600 K)
Perovskites
MAPbBr3 (300 K)
Metal dichalcogenides
bulk MoS2 (300 K)
(b)
(c)
(d)
Reference
MACE-MP-0 (20.2%)
GRACE-2L-MPtrj (28.0%)
Tier 1 mean VDOS error: 18.4%
MACE-MPA-0 (19.8%)
orb-v2 (24.5%)
Tier 2 mean VDOS error: 13.2%
PET-OMAT-XL (15.7%)
MatterSim-v1.0.0-5M (22.0%)
Tier 3 mean VDOS error: 12.4%
UMA-S-P1 (turbo) (11.3%)
UMA-M-P1 (turbo) (19.0%)
Tier 4 mean VDOS error: 10.0%
MACE-MP-0 (25.2%)
GRACE-2L-MPtrj (27.2%)
Tier 1 mean VDOS error: 20.1%
MACE-MPA-0 (27.0%)
orb-v2 (27.7%)
Tier 2 mean VDOS error: 17.5%
PET-OAM-XL (19.1%)
MatterSim-v1.0.0-5M (30.2%)
Tier 3 mean VDOS error: 16.0%
UMA-M-P1 (compile) (17.4%)
MACE-MH-1-OMAT (23.0%)
Tier 4 mean VDOS error: 14.4%
GRACE-2L-MPtrj (63.8%)
MACE-MP-0 (72.2%)
Tier 1 mean VDOS error: 56.6%
MACE-MPA-0 (66.9%)
orb-v2 (86.7%)
Tier 2 mean VDOS error: 52.3%
PET-OAM-XL (38.5%)
orb-v3-direct-20-omat (70.8%)
Tier 3 mean VDOS error: 40.5%
UMA-M-P1 (compile) (45.5%)
MACE-MH-1-OMAT (70.7%)
Tier 4 mean VDOS error: 35.1%
VDOS
Energy [eV]
Tier 1
Tier 2
Tier 3
Tier 4
Metal alloys
bulk Pt3Co (300 K)
Metal-water interfaces
Pt(111) + 24 H2O (380 K)
Hydrogen
H (1050 K)
(e)
(f)
(g)
GRACE-2L-MPtrj (27.2%)
MACE-MP-0 (39.9%)
Tier 1 mean VDOS error: 16.0%
MACE-MPA-0 (24.6%)
orb-v2 (29.1%)
Tier 2 mean VDOS error: 13.3%
PET-OAM-XL (21.8%)
MatterSim-v1.0.0-5M (27.3%)
Tier 3 mean VDOS error: 12.9%
MACE-MH-1-OMAT (26.0%)
UMA-S-P1 (turbo) (35.4%)
Tier 4 mean VDOS error: 14.0%
GRACE-2L-MPtrj (39.3%)
MACE-MP-0 (40.3%)
Tier 1 mean VDOS error: 39.8%
MACE-MPA-0 (30.1%)
orb-v2 (30.5%)
Tier 2 mean VDOS error: 30.3%
orb-v3-direct-20-omat (22.1%)
NequIP-OAM-L (33.0%)
Tier 3 mean VDOS error: 28.6%
UMA-M-P1 (turbo) (29.9%)
UMA-M-P1 (compile) (33.2%)
Tier 4 mean VDOS error: 31.9%
Tier 1 mean VDOS error: 100.0%
orb-v2 (46.9%)
Tier 2 mean VDOS error: 73.4%
PET-OMAT-XL (5.7%)
MatterSim-v1.0.0-5M (52.1%)
Tier 3 mean VDOS error: 41.2%
UMA-S-P1 (compile) (35.6%)
UMA-S-P1 (turbo) (35.7%)
Tier 4 mean VDOS error: 74.3%